Phase (1) - Mathematical Modeling of the Cantilever Beam

The first phase of the project involves deriving the mathematical model of the cantilever beam using Hamilton's principle.

(1) Introduction:

A cantilever beam is a fundamental structural element fixed at one end(x=0) and free at the other (x= L).

For the purpose of this derivation, consider a beam of length L, cross-sectional area A, mass density ρ, and flexural rigidity EI. As discussed earlier, to model the natural flexibility of the fixed support of the cantilever beam, a rotational spring of stiffness kΘ is considered. An infinitesimal element of length 'dx' is located at a distance 'x' from the fixed support as shown in the figure:

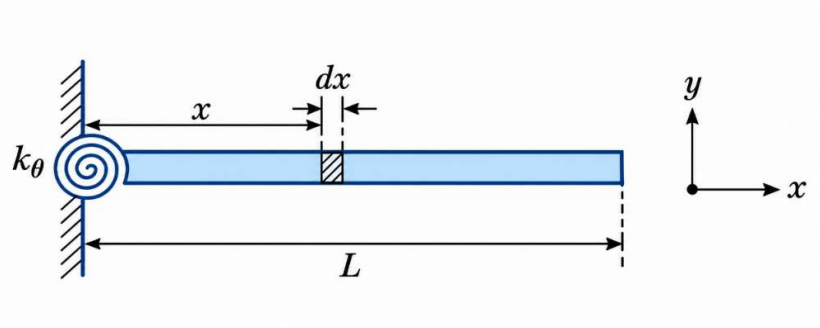


Fig. 1 Schematic diagram of cantilever beam setup

(2). Energy formulations:

2.1 Hamilton's Principle

Hamilton's principle states that the actual motion of a mechanical system minimizes the dynamic action integral between two fixed times t_1 and t_2:

$$\int_{t1}^{t2}(\delta U - \delta T)\, dt = 0 \quad (1)$$

2.2 Strain Energy (U)

The strain energy stored in the beam due to bending moment $M_x = EI\ \partial^2v/\partial x^2$ is given by:

$$U = \int_0^L \frac{M_x^2}{2EI}dx = \int_0^L \frac{1}{2E}\left(EI\frac{\partial^2 v}{\partial^2 x}\right)^2 dx \quad (2)$$

Simplifying equation (2):

$$U = \frac{EI}{2}\int_0^L \frac{\partial^2 v^2}{\partial^2 x}\ dx \quad (3)$$

Taking the variation of the strain energy yields:

$$\delta U = EI\int_0^L \frac{\partial^2 v}{\partial^2 x}\delta\left(\frac{\partial^2 v}{\partial^2 x}\right) dx \quad (4)$$

$$U_{spring} = \frac{1}{2}k_\theta\theta^2|_{x=0} \rightarrow U_{spring} = \frac{1}{2}k_\theta\left(\frac{\partial v}{\partial x}\right)^2|x=0 \quad (5)$$

$$U = U_{beam} + U_{spring}$$

$$U = \frac{EI}{2}\int_0^L\left(\frac{\partial^2 v}{\partial x^2}\right)^2 dx + \frac{1}{2}k_\theta\left(\frac{\partial v}{\partial x}\right)^2 \quad (6)$$

2.3 Kinetic Energy (T)

The continuous kinetic energy of the beam element with mass element $dm = \rho A\ dx$ and lateral velocity $\partial v/\partial t$ is:

$$T = \int_0^L \tfrac{1}{2}(\rho A\ dx)\ (\partial v/\partial t)^2 = \frac{\rho A}{2}\int_0^L (\partial v/\partial t)^2\ dx \quad (7)$$

The variation of kinetic energy is:

$$\delta T = \rho A\int_0^L (\partial v/\partial t)\ \delta(\partial v/\partial t)\ dx \quad (8)$$

(3). Substituting above equations in Hamilton Equation

Putting obtained energy expressions in the Hamilton's equation, we get:

$$\int_{t_1}^{t_2}\left[\ EI\int_0^L \frac{\partial^2 v}{\partial x^2}\cdot\delta\left(\frac{\partial^2 v}{\partial x^2}\right)dx + k\theta_\theta\left(\frac{\partial v}{\partial x}\right)_{|x=0}\cdot\delta\left(\left(\frac{\partial v}{\partial x}\right)_{|x=0}\right) - \rho A\int_0^L \frac{\partial v}{\partial t}\cdot\delta\left(\frac{\partial v}{\partial t}\right)dx\ \right]dt = 0 \quad (9)$$

From previous knowledge, we know that:

$$\int_{t_1}^{t_2} (\rho A \int_0^L \frac{\partial v}{\partial t}\ \delta(\frac{\partial v}{\partial t})\, dx)\, dt\ =\ -\int_{t_1}^{t_2} (\rho A \int_0^L \partial^2 v\ \delta v\ /\ \partial t^2\, dx)\, dt \qquad (10)$$

So, equation (9) becomes:

$$\int_{t_1}^{t_2} [\, EI \int_0^L \frac{\partial^2 v}{\partial x^2}\ \delta(\frac{\partial^2 v}{\partial x^2})\, dx\ +\ k\theta_\theta\ \frac{\partial v}{\partial x}|x=0\ \delta(\frac{\partial v}{\partial x}|x=0)\ +\ \rho A \int_0^L \frac{\partial^2 v}{\partial t^2}\ \delta v\, dx\,]\, dt\ =\ 0$$

Since the spatial equation (inside the integral) must hold true for any arbitrary variation over time, the overall time integral can be removed.

Thus, removing the time integral we get:

$$EI \int_0^L \frac{\partial^2 v}{\partial x^2}\ \delta(\frac{\partial^2 v}{\partial x^2})\, dx\ +\ k\theta_\theta\ \frac{\partial v}{\partial x}|x=0\ \delta(\frac{\partial v}{\partial x}|x=0)\ +\ \rho A \int_0^L \frac{\partial^2 v}{\partial t^2}\ \delta v\, dx\ =0 \quad (11)$$

(3). Non-Dimensional beam domain

$\xi\ =\ \frac{x}{L}$

x = L. ξ → dx = L. (d ξ) (12)

Putting (12) into (11) we get:

$$EI \int_0^{1/L^3} \frac{\partial^2 v}{\partial \xi^2}\ \delta(\frac{\partial^2 v}{\partial \xi^2})\, d\xi\ +\ \frac{k\theta_\theta}{L^2}\ \frac{\partial v}{\partial \xi}|\xi=0\ \delta(\frac{\partial v}{\partial \xi}|\xi=0)\ +\ \rho AL \int_0^{1/L} \frac{\partial^2 v}{\partial t^2}\ \delta v\, d\xi\ =\ 0 \qquad (13)$$

Now, lets approximate the dynamic displacement field as:

$$v(\xi,t)\ =\ \sum_{j=1}^{n} C_j\ \varphi_j(\xi)\ e^{i\omega t} \qquad (14)$$

under the assumption that the displacement field is separable in space and time.

In equation (14):

C_j are known coefficients

φ_j is the shape function

ω is the natural frequency

In the approximate field, φ(ξ) is the shape function to be chosen satisfying the essential boundary conditions

$$\varphi_j(\xi) = sin(j\xi)$$

where j = 1, 2, 3, …

$$At\ \xi = 0:\ \varphi_j(0) = sin(0) = 0$$

$$\varphi_j'(0) = j\ cos(0) = j \neq 0$$

$$At\ \xi = 1:\ \varphi_j(1) = sin(j) \neq 0$$

$$\varphi_j'(1) = j\ cos(j) \neq 0$$

$$v(\xi, t) = \sum_{j=1}^{n} C_j\ \varphi_j(\xi)\ e^{i\omega t} \qquad (14)$$

$$\frac{\partial^2 v}{\partial \xi^2} = \sum_{j=1}^{n} C_j\ \frac{d^2\varphi_j}{d\xi^2}\ e^{i\omega t}$$

$$\frac{\partial^2 v}{\partial t^2} = \sum_{j=1}^{n} C_j\ \varphi_j\ (-\omega^2)\ e^{i\omega t}$$

Substituting these fields into the variation equation, we get:

$$\left[\frac{EI}{L^3} \int_0^1 \sum_{j=1}^{n} C_j\ \frac{d^2\varphi_j}{d\xi^2}\ e^{\ i\omega t}\ \ \delta\left(\sum_{j=1}^{n} C_j\ \frac{d^2\varphi_j}{d\xi^2}\ e^{\ i\omega t}\right) d\xi\right] + \left[\frac{k\theta_\theta}{L^2}\ \frac{\partial v}{\partial \xi}|\xi = 0\ \ \delta\left(\frac{\partial v}{\partial \xi}\right)|\xi = 0\right] +$$

$$\left[\rho AL \int_0^1 \sum_{j=1}^{n} C_j\ \varphi_j\ (-\omega^2) e^{\ i\omega t}\ \ \delta v\ d\xi\right] = 0 \qquad (15)$$

$$\frac{\partial v}{\partial \xi} = \sum_{j=1}^{n} C_j\ \frac{d\varphi_j}{d\xi}\ e^{i\omega t}$$

$$\frac{\partial v}{\partial \xi}|\xi = 0 = \sum_{j=1}^{n} C_j\ \frac{d\varphi_j}{d\xi}|\xi = 0\ e^{i\omega t}$$

Let $\delta v = \varphi_k$

$$\delta\left(\sum_{j=1}^{n} C_j\ \frac{d^2\varphi_j}{d\xi^2}\ e^{i\omega t}\right) = \sum_{k=1}^{n} \frac{d^2\varphi_k}{d\xi^2} \qquad (16)$$

$$\delta\left(\sum_{j=1}^{n} C_j\ \frac{d\varphi_j}{d\xi}|\xi = 0\ e^{i\omega t}\right) = \sum_{k=1}^{n} \frac{d\varphi_k}{d\xi}|\xi = 0 \qquad (17)$$

Putting equations (16) and (17) into equation (15), we get:

$$[EI/L^3 \int_0^1 \sum_{j=1}^{n} C_j \frac{d^2\varphi_j}{d\xi^2} \sum_{k=1}^{n} \frac{d^2\varphi_k}{d\xi^2} d\xi + k_\theta/L^2 \sum_{j=1}^{n} C_j \frac{d\varphi_j}{d\xi} |\xi_0 \sum_{k=1}^{n} \frac{d\varphi_k}{d\xi} |\xi_0 - \omega^2 \rho A L \left[\int_0^1 \sum_{j=1}^{n} C_j \varphi_j \varphi_k d\xi\right] e^{i\omega t} = 0] \quad (18)$$

For arbitrary time t, equation (18) becomes:

$$[\frac{EI}{L^3} \int_0^1 \sum_{j=1}^{n} C_j \frac{d^2\varphi_j}{d\xi^2} \sum_{k=1}^{n} \frac{d^2\varphi_k}{d\xi^2} d\xi + \frac{k_\theta}{L^2} \sum_{j=1}^{n} C_j \frac{d\varphi_j}{d\xi} |\xi_0 \sum_{k=1}^{n} \frac{d\varphi_k}{d\xi} |\xi_0 - \omega^2 \rho A L \int_0^1 \sum_{j=1}^{n} C_j \varphi_j \varphi_k d\xi \; e^{i\omega t}] C_j = 0$$

$$\sum_{j=1}^{n}\sum_{k=1}^{n}\left[(\frac{EI}{L^3}\int_0^1 \frac{d^2\varphi_j}{d\xi^2}\frac{d^2\varphi_k}{d\xi^2} d\xi + \frac{k_\theta}{L^2}\frac{d\varphi_j}{d\xi}|\xi_0 \frac{d\varphi_k}{d\xi}|\xi_0)\right] - \omega^2 \int_0^1 \rho A L \varphi_j \varphi_k d\xi \; (CC_j) = 0 \quad (19)$$

Equation (19) can be written as:

$$([K] - \omega^2[M])\{C_i\} = \{0\} \quad (20)$$

Where:

Stiffness matrix:

$$[K] = \sum_{j=1}^{n}\sum_{k=1}^{n}(\frac{EI}{L^3}\int_0^1 \frac{d^2\varphi_j}{d\xi^2}\frac{d^2\varphi_k}{d\xi^2} d\xi + \frac{k\theta_\theta}{L^2}\frac{d\varphi_j}{d\xi}|\xi = 0 \frac{d\varphi_k}{d\xi}|\xi = 0)$$

Mass matrix:

$$[M] = \sum_{j=1}^{n}\sum_{k=1}^{n} \rho AL \int_0^1 \varphi_j \varphi_k d\xi$$

Phase (2) Digital Modelling of the Cantilever Beam setup